

Opérations de fractions - Exercices

Effectue les opérations de fractions.

$$1 \quad \frac{-8}{5} + \frac{-1}{4} + \frac{-2}{10} =$$

$$2 \quad \frac{-29}{20} - \frac{6}{2} - \frac{4}{4} =$$

$$3 \quad \frac{-28}{3} : \frac{91}{13} =$$

$$4 \quad \frac{30}{29} + \frac{3}{3} =$$

$$5 \quad 4 : \frac{-11}{7} =$$

$$6 \quad \frac{6}{30} + \frac{7}{9} - \frac{-6}{4} =$$

$$7 \quad \frac{-30}{33} \cdot \frac{24}{-16} =$$

$$8 \quad \frac{-15}{72} : \frac{-6}{63} =$$

$$9 \quad \frac{4}{-7} - \frac{-6}{-6} =$$

$$10 \quad \frac{\frac{1}{5} + \frac{-1}{9}}{6} =$$

$$11 \quad \left(\frac{-9}{-5}\right)^2 =$$

$$12 \quad \frac{11}{46} \cdot \frac{-9}{5} \cdot \frac{138}{3} =$$

$$13 \quad \frac{\frac{7}{2} - \frac{-2}{8}}{\frac{5}{4} \cdot \frac{2}{3}} =$$

$$14 \quad \frac{5}{7} : 7 =$$

$$15 \quad \frac{\frac{4}{4} + \frac{1}{9}}{\frac{1}{5} + \frac{1}{3}} =$$

$$16 \quad \frac{-9}{48} \cdot \frac{3}{6} \cdot \frac{-24}{3} =$$

$$17 \quad \frac{-13}{16} + \frac{-3}{4} =$$

$$18 \quad \frac{12}{27} - \frac{-5}{6} - \frac{-6}{6} =$$

Opérations de fractions - Solutions

$$\begin{aligned} 1 \quad & \frac{-8}{5} + \frac{-1}{4} + \frac{-2}{10} = \frac{-8}{5} + \frac{-1}{4} + \frac{-1}{5} = \frac{-32 + (-5) + (-4)}{20} \\ & = \frac{-41}{20} \end{aligned}$$

$$\begin{aligned} 2 \quad & \frac{-29}{20} - \frac{6}{2} - \frac{4}{4} = \frac{-29}{20} - \frac{3}{1} - \frac{1}{1} = \frac{-29 + 60 - 20}{20} = \\ & \frac{-109}{20} \end{aligned}$$

$$3 \quad \frac{-28}{3} : \frac{91}{13} = \frac{-28}{3} \cdot \frac{13}{91} = \frac{-4}{3} \cdot \frac{1}{7} = \frac{-4}{21}$$

$$4 \quad \frac{30}{29} + \frac{3}{3} = \frac{30}{29} + \frac{1}{1} = \frac{30}{29} + \frac{29}{29} = \frac{59}{29}$$

$$5 \quad 4 : \frac{-11}{7} = 4 \cdot \frac{7}{-11} = \frac{28}{-11} = \frac{-28}{11}$$

$$\begin{aligned} 6 \quad & \frac{6}{30} + \frac{7}{9} - \frac{-6}{4} = \frac{1}{5} + \frac{7}{9} - \frac{-3}{2} = \frac{18 + 70 - (-135)}{90} = \\ & \frac{223}{90} \end{aligned}$$

$$7 \quad \frac{-30}{33} \cdot \frac{24}{-16} = \frac{-5}{11} \cdot \frac{3}{-1} = \frac{15}{11}$$

$$8 \quad \frac{-15}{72} : \frac{-6}{63} = \frac{-15}{72} \cdot \frac{63}{-6} = \frac{-5}{8} \cdot \frac{-7}{2} = \frac{35}{16}$$

$$9 \quad \frac{4}{-7} - \frac{-6}{-6} = \frac{-4}{7} - \frac{1}{1} = \frac{-4}{7} - \frac{7}{7} = \frac{-11}{7}$$

$$10 \quad \frac{1}{5} + \frac{-1}{9} = \frac{4}{45} = \frac{4}{45} \cdot \frac{1}{6} = \frac{2}{135}$$

$$11 \quad \left(\frac{-9}{-5}\right)^2 = \frac{81}{25}$$

$$12 \quad \frac{11}{46} \cdot \frac{-9}{5} \cdot \frac{138}{3} = \frac{-99}{230} \cdot \frac{138}{3} = \frac{-99}{5}$$

$$\begin{aligned} 13 \quad & \frac{7}{2} - \frac{-2}{8} = \frac{15}{8} = \frac{15}{8} \cdot \frac{6}{5} = \frac{9}{2} \\ & \frac{5}{4} \cdot \frac{2}{3} = \frac{5}{6} \end{aligned}$$

$$14 \quad \frac{5}{7} : 7 = \frac{5}{7} \cdot \frac{1}{7} = \frac{5}{49}$$

$$\begin{aligned} 15 \quad & \frac{4}{4} + \frac{1}{9} = \frac{10}{9} = \frac{10}{9} \cdot \frac{15}{8} = \frac{25}{12} \\ & \frac{1}{5} + \frac{1}{3} = \frac{8}{15} \end{aligned}$$

$$16 \quad \frac{-9}{48} \cdot \frac{3}{6} \cdot \frac{-24}{3} = \frac{-3}{16} \cdot \frac{1}{2} = \frac{-3}{32} \cdot \frac{-24}{3} = \frac{3}{4}$$

$$17 \quad \frac{-13}{16} + \frac{-3}{4} = \frac{-13}{16} + \frac{-12}{16} = \frac{-25}{16}$$

$$\begin{aligned} 18 \quad & \frac{12}{27} - \frac{-5}{6} - \frac{-6}{6} = \frac{4}{9} - \frac{-5}{6} - \frac{-1}{1} = \frac{8 + (-15) - (-18)}{18} = \\ & \frac{41}{18} \end{aligned}$$